TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 1

CHAPTER 5: ALKALI AND ALKALINE EARTH METALS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 1
Chapter Name:	Alkali and Alkaline Earth Metals

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

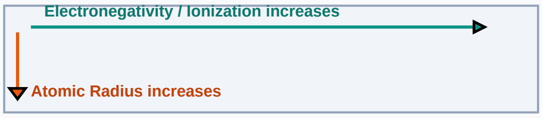
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Alkali Metals	Group 1 elements (Li, Na, K, Rb, Cs, Fr) having a single s valence electron, forming strongly alkaline oxides.
Alkaline Earth Metals	Group 2 elements (Be, Mg, Ca, Sr, Ba, Ra) having two s valence electrons, forming basic hydroxides.
Diagonal Relationship	The similarity in properties of certain light elements in the second period with adjacent elements in the third period (e.g. Li-Mg, Be-Al).

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

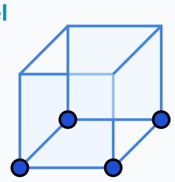
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Simple Cubic Lattice System

Essential Formulas, Laws & Identities:

- Plaster of Paris: $CaSO_4 . 0.5H_2O$
- Gypsum dehydration: $CaSO_4 . 2H_2O \rightarrow CaSO_4 . 0.5H_2O + 1.5H_2O$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

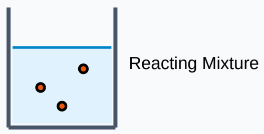
Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Why does Lithium show a diagonal relationship with Magnesium?

Step-by-step Solution:
Lithium and Magnesium have similar ionic sizes (Li^+ = 0.76A, Mg^{2+} = 0.72A) and similar electronegativities (Li = 1.0, Mg = 1.2), yielding identical polarizing powers.

Titration Beaker & Solute Precipitation



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Describe the industrial Solvay Process for the preparation of Sodium Carbonate (Na_2CO_3) with all chemical equations.

Analytical Solution Strategy:

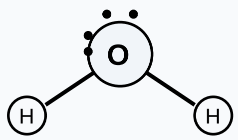
Process involves passing CO_2 into ammoniacal brine:

$NH_3 + H_2O + CO_2 \rightarrow NH_4HCO_3$.

$NH_4HCO_3 + NaCl \rightarrow NaHCO_3 + NH_4Cl$.

$NaHCO_3$ is heated to yield sodium carbonate: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Explain why Group 1 metals impart characteristic colors to Bunsen burner flames.

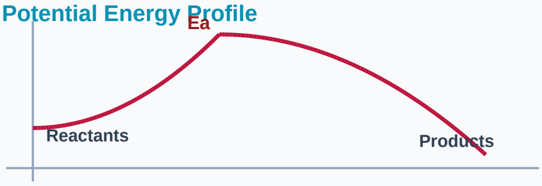
Q2 (2-Mark): State the biological importance of Calcium and Magnesium ions.

Q3 (2-Mark): Describe the preparation and uses of Quicklime (CaO).

Q4 (5-Mark): Why is beryllium anomalous in Group 2?

Q5 (5-Mark): What is dead burnt plaster? How is it formed?

Chemical Reaction Potential Energy Profile



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Construction Materials

Cement formulations depend on gypsum additions to regulate setting speeds during pouring.

Application Case: Medicine

Epsom salt ($MgSO_4.7H_2O$) and milk of magnesia are used as laxatives and antacids.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C_6H_6) Resonance

Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Electrochemical Galvanic Cell System